

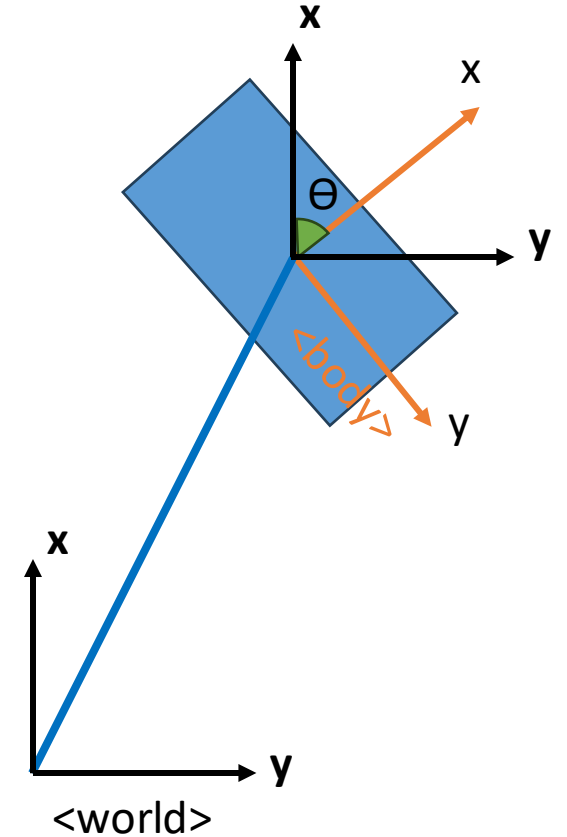
CLMVS – Solution of lab2

Controllability

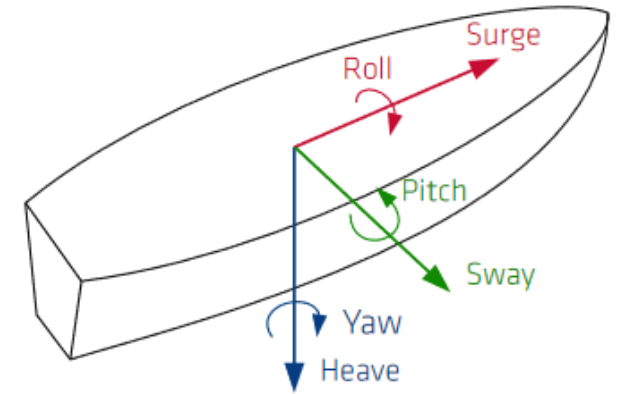
$$\begin{aligned} m\ddot{x} &= -\alpha\dot{x} + F_x \\ m\ddot{y} &= -\beta\dot{y} + F_y \\ J\dot{w} &= -\gamma w + N_z \end{aligned}$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\frac{d}{dt}x = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix} \begin{bmatrix} r_x \\ r_y \\ \theta \\ \dot{r}_x \\ \dot{r}_y \\ w \end{bmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/J \end{pmatrix} \begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix}$$

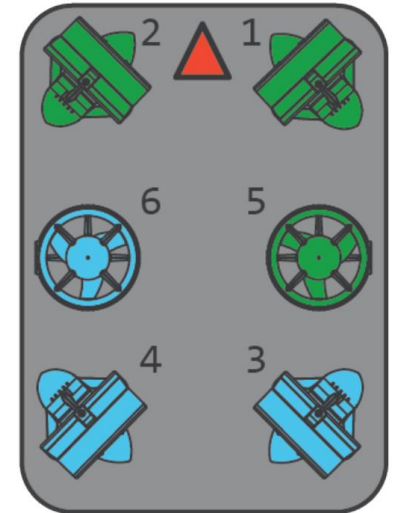


Thruster allocation matrix



Thruster allocation matrix (TAM)

	T1	T2	T3	T4	T5	T6
Surge	0.7071	0.7071	-0.7071	-0.7071	0	0
Sway	-0.7071	0.7071	-0.7071	0.7071	0	0
Heave	0	0	0	0	-1	1
Roll	0	0	0	0	0.218	0.218
Pitch	0	0	0	0	0.12	-0.12
Yaw	-0.1888	0.1888	0.1888	-0.1888	0	0

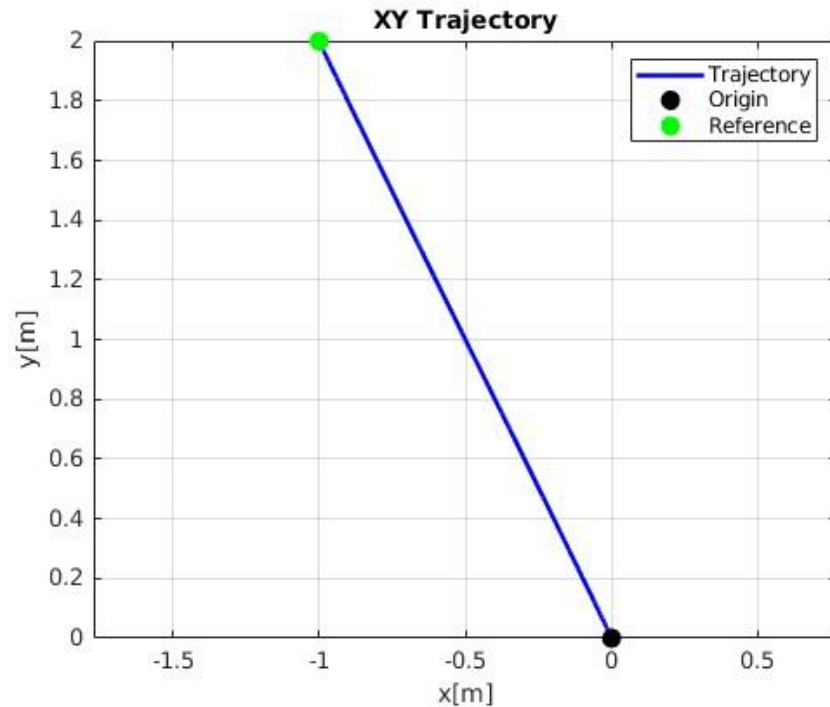


CLMVS – Solution of lab2

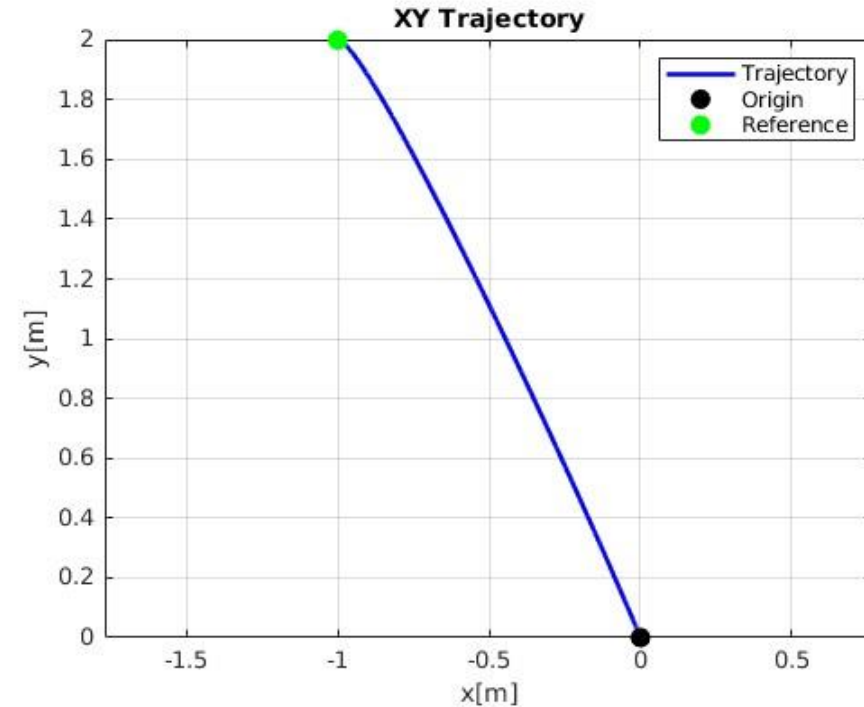
Controllability

3 cases:

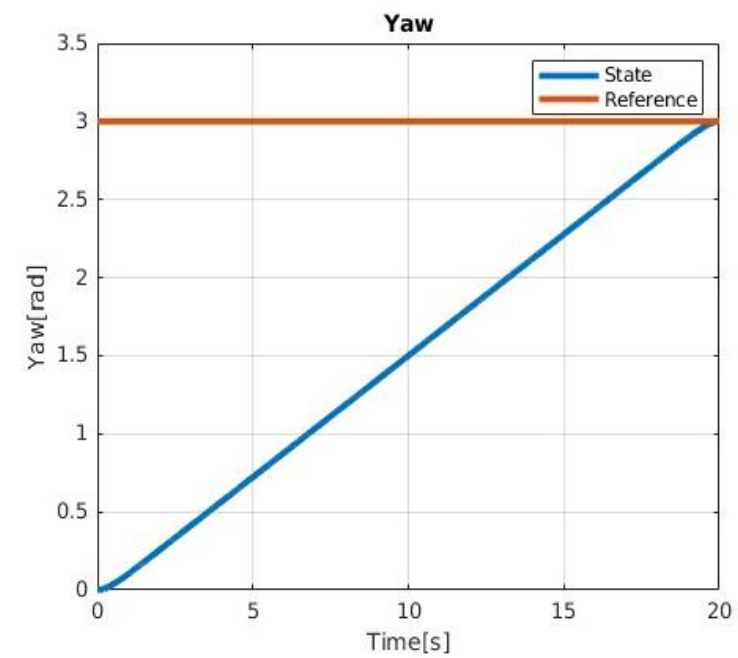
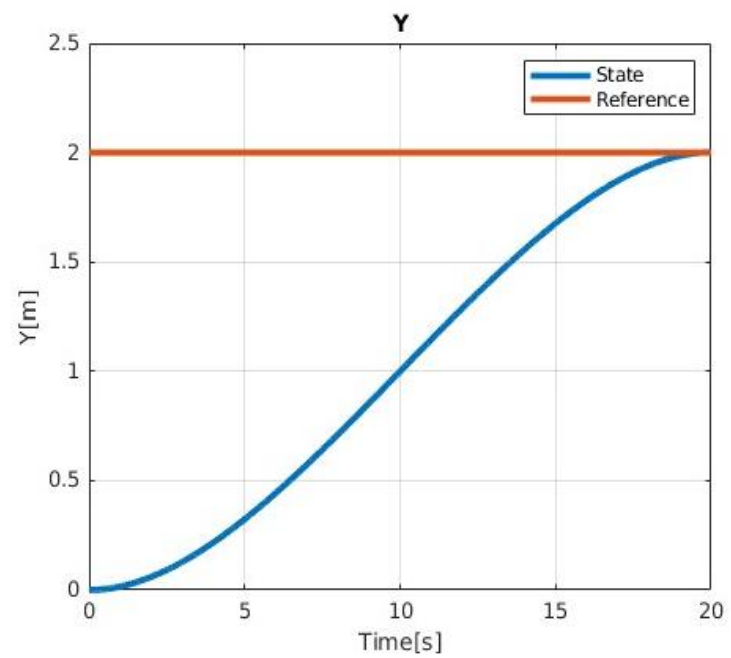
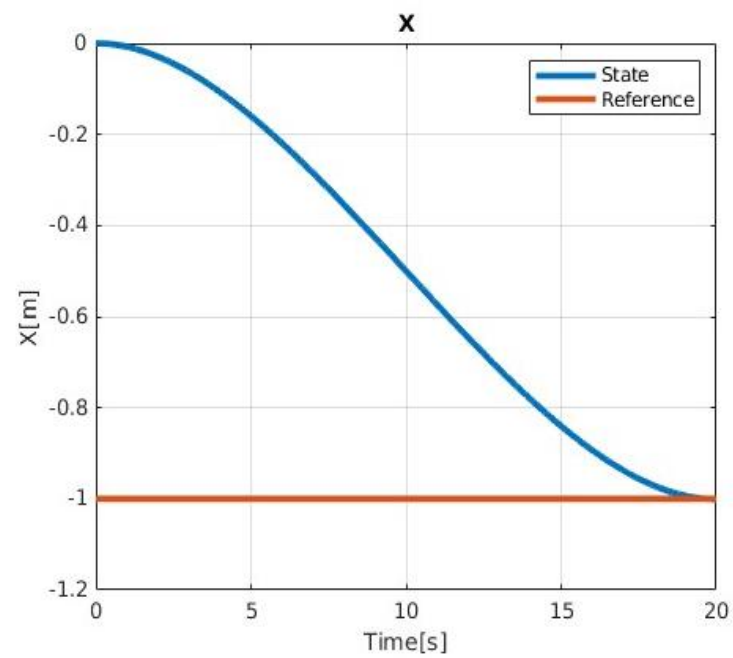
1. **All thrusters enabled:** fully controllable
2. **Thruster 2 disabled:** fully controllable (thanks to redundancy)



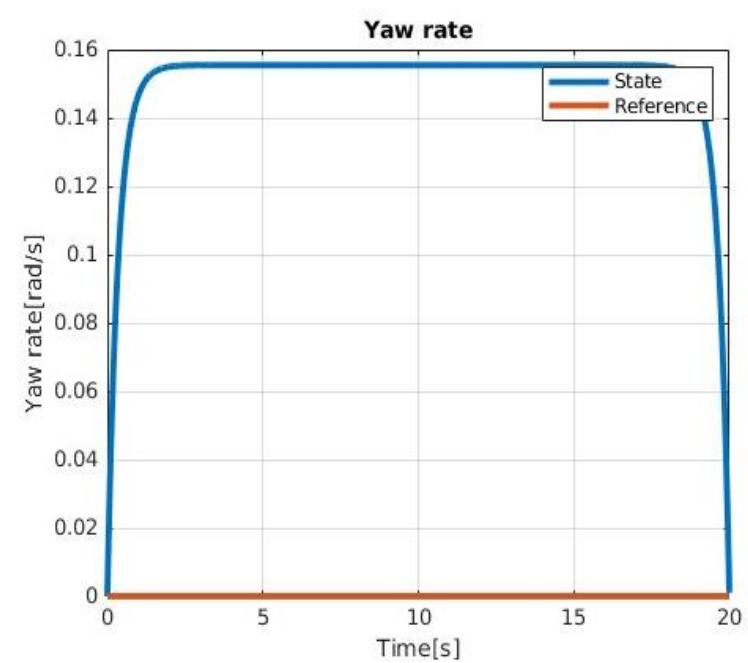
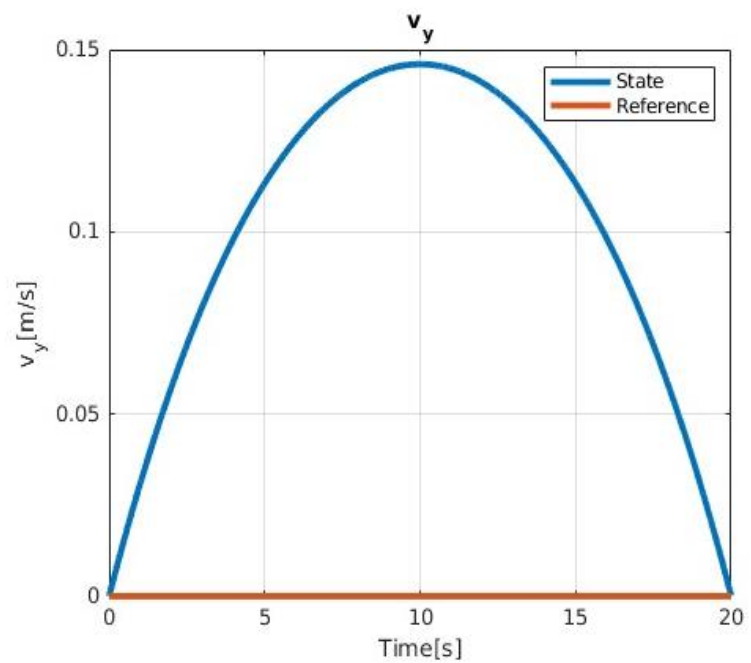
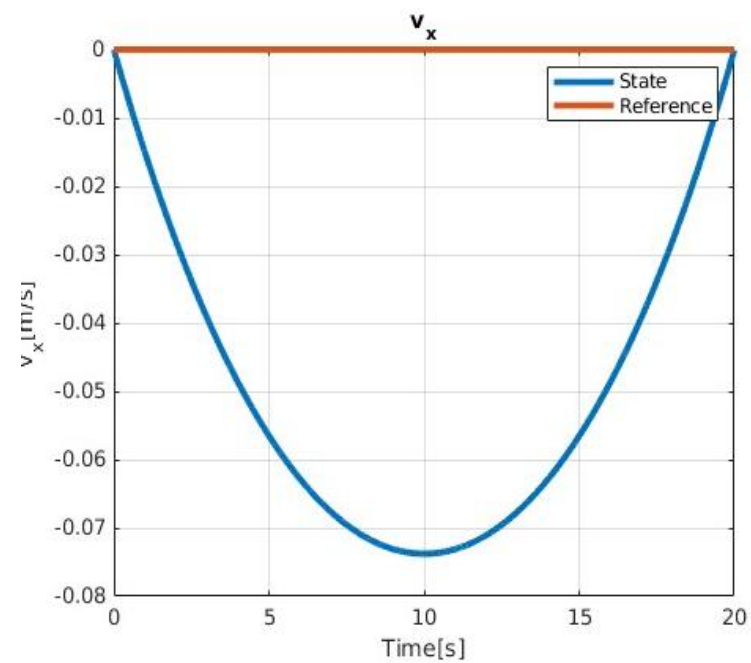
Case 1

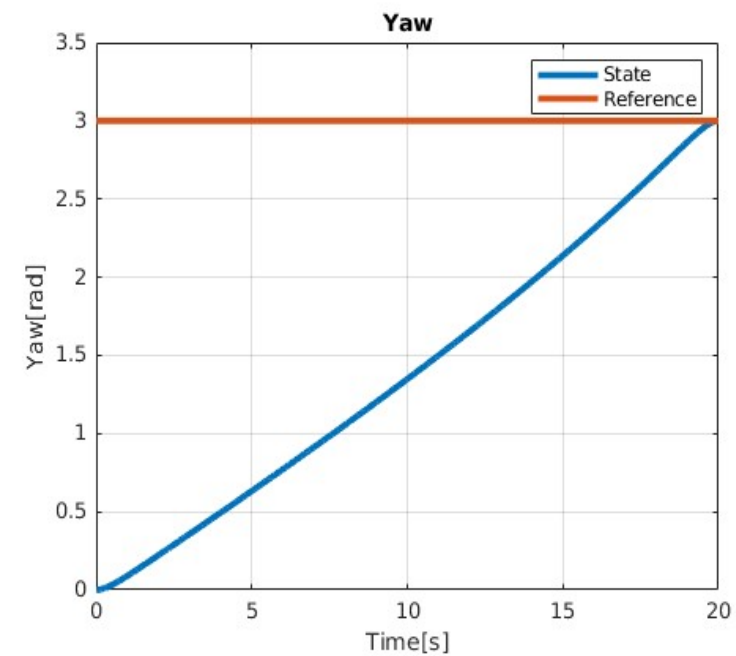
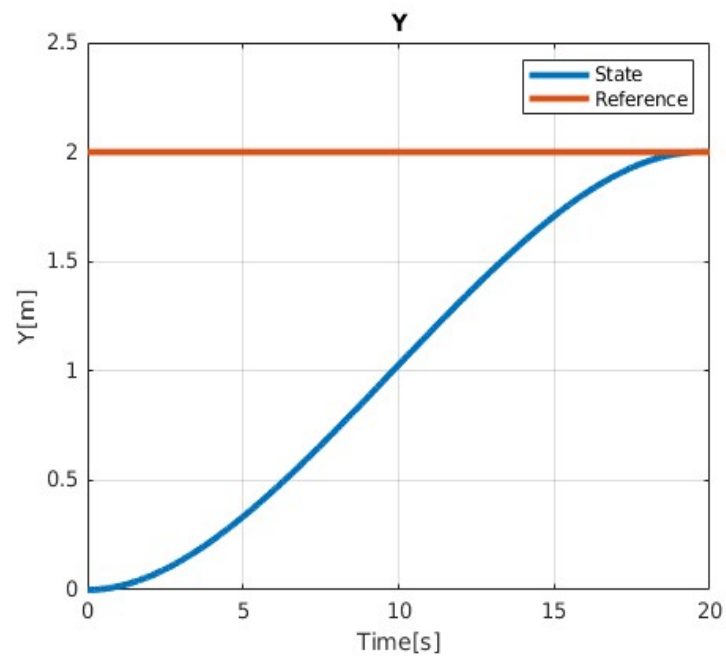
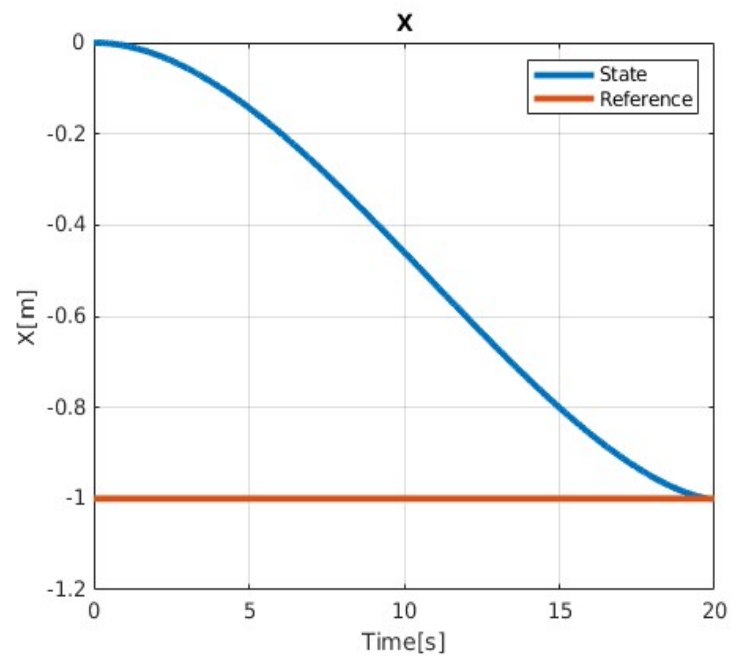


Case 2

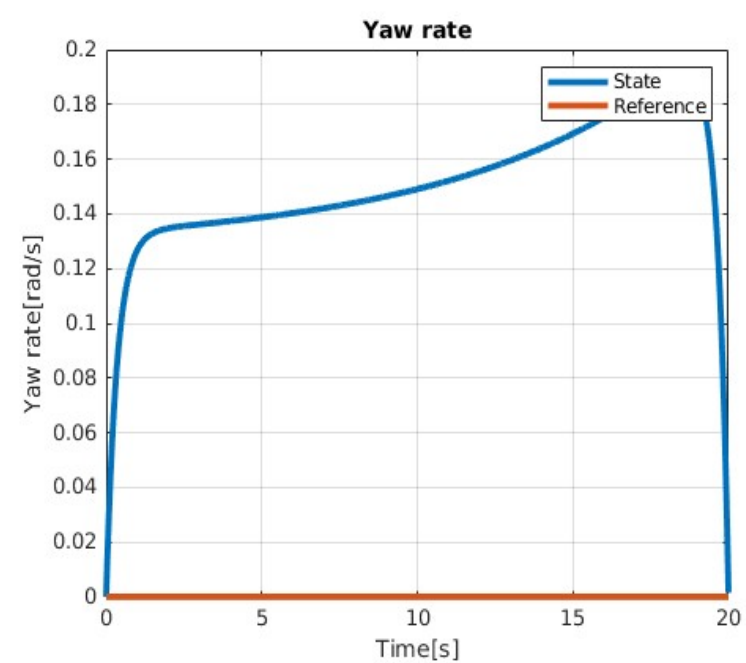
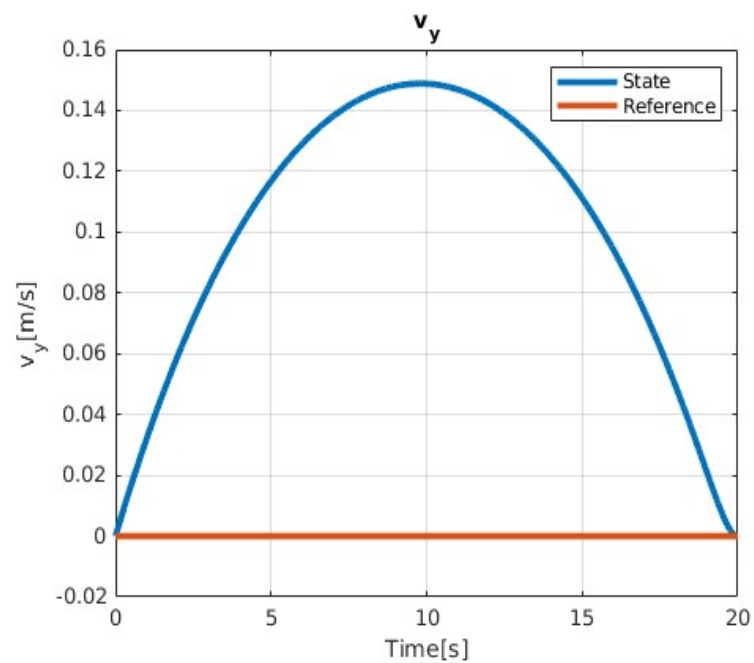
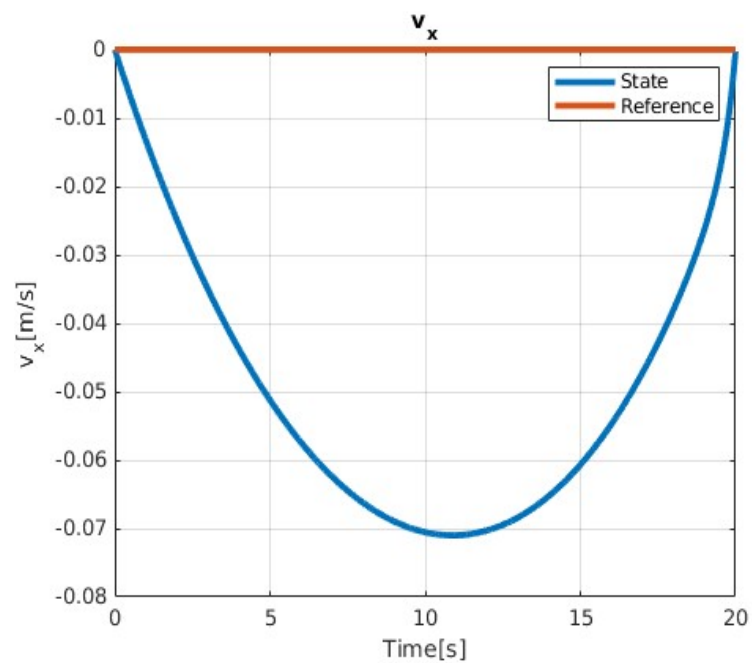


Case 1





Case 2



CLMVS – Solution of lab2

Controllability

3. Thrusters 2,3 disabled: not fully controllable ($\text{rank}(\mathbf{G}) = 5$)
What is controllable in this case?

Take the SVD of the controllability Gramian $\mathbf{G}$:

$$[\mathbf{U}, \mathbf{D}, \mathbf{V}] = \text{svd}(\mathbf{G})$$

The first $\text{rank}(\mathbf{G})$ columns of $\mathbf{U}$ are an orthonormal basis of the controllable subspace. Let's analyze $\mathbf{U}$...

CLMVS – Solution of lab2

Controllability

U =

controllable subspace

-0.7551	-0.0000	0.0000	-0.1656	0.6302	0.0728
0.6539	-0.0000	0.0000	-0.1214	0.7409	0.0936
0	0.9996	0.0277	0	0	0
-0.0373	0.0000	0.0000	0.7085	0.0603	0.7021
0.0284	0.0000	0.0000	-0.6752	-0.2244	0.7021
-0.0000	0.0277	-0.9996	0.0000	0.0000	0.0000

- Columns 2 and 3 only have non-zero components on the third and sixth row (θ , ω)
- Moreover, they are not proportional and therefore are linearly independent
- This means that whole rotation subspace is controllable, that is you can rotate by any specified amount and with any target final angular velocity

CLMVS – Solution of lab2

Controllability

$U =$

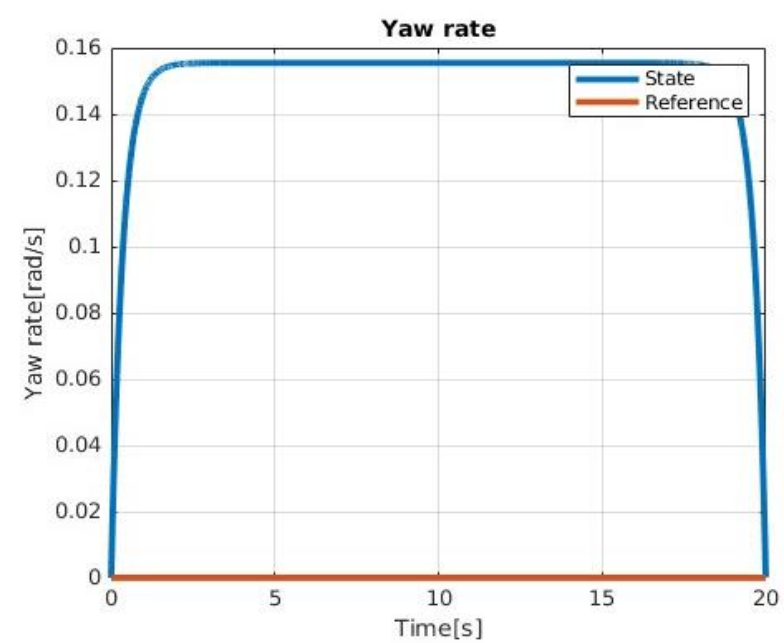
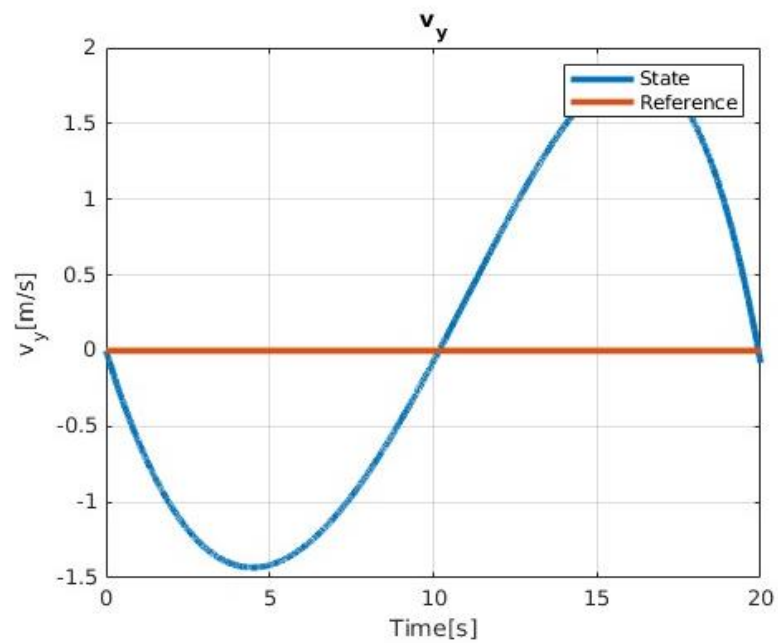
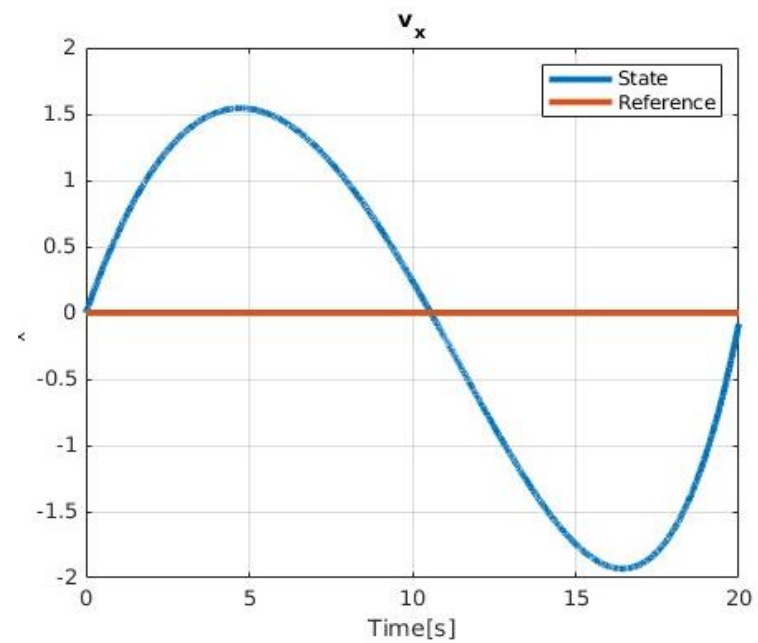
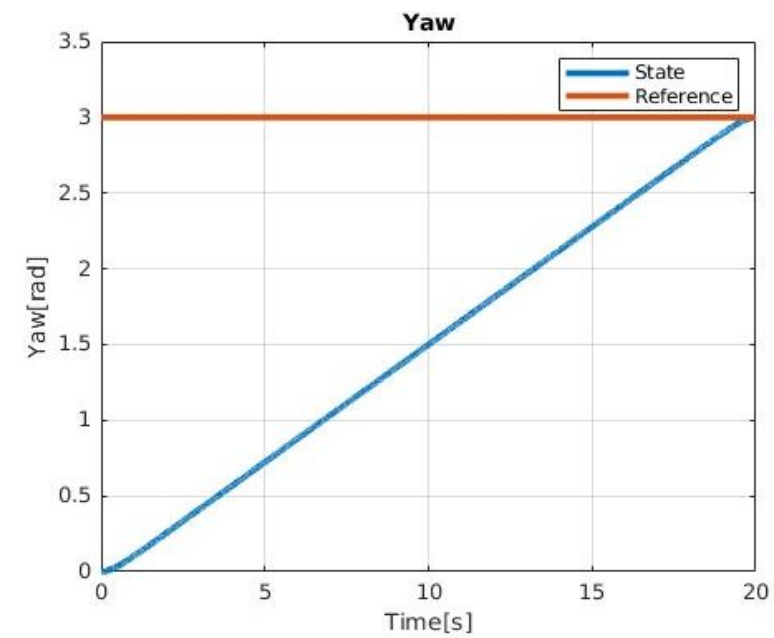
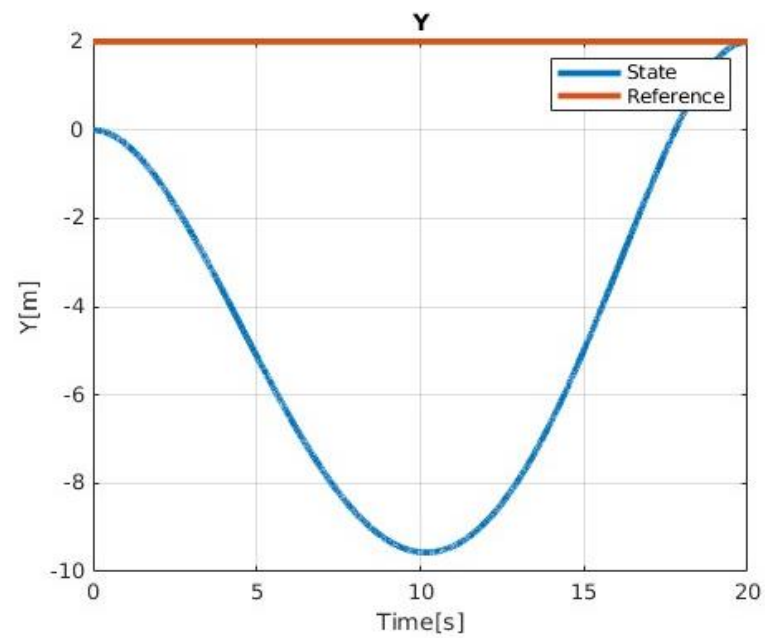
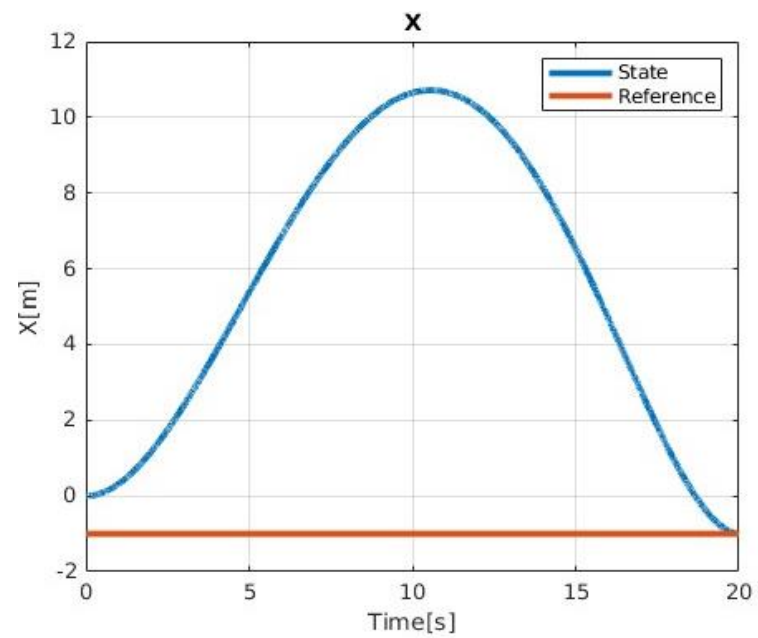
controllable subspace					
-0.7551	-0.0000	0.0000	-0.1656	0.6302	0.0728
0.6539	-0.0000	0.0000	-0.1214	0.7409	0.0936
0	0.9996	0.0277	0	0	0
-0.0373	0.0000	0.0000	0.7085	0.0603	0.7021
0.0284	0.0000	0.0000	-0.6752	-0.2244	0.7021
-0.0000	0.0277	-0.9996	0.0000	0.0000	0.0000

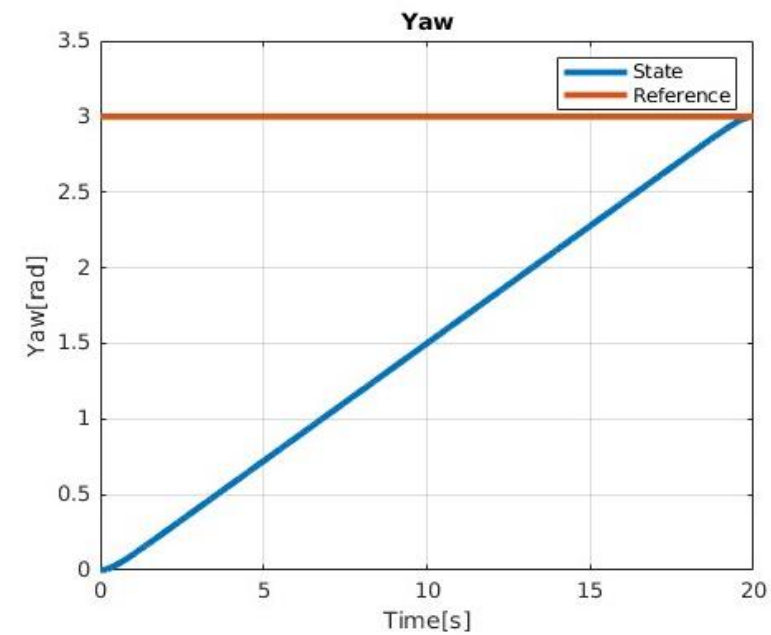
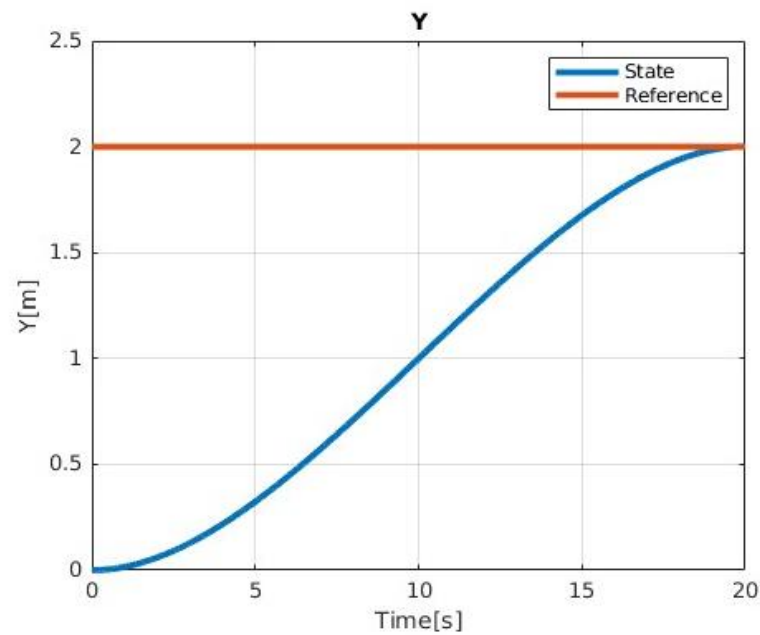
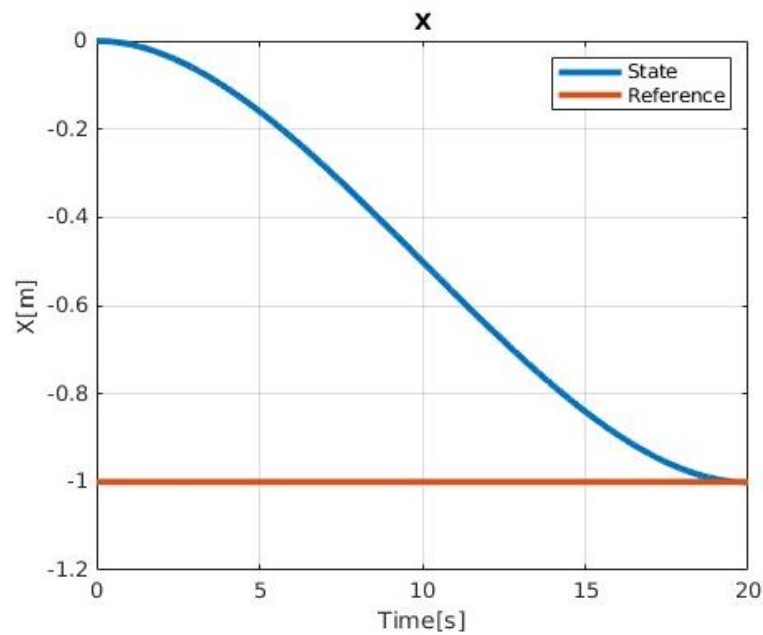
- Columns 1, 4 and 5 are linearly independent (check!)
- However, the subspace $(x, y, \dot{x}, \dot{y})$ is 4-dimensional
- 3 vectors alone are not sufficient to span a 4-dimensional vector space
- This means that you can reach any final position, but not with arbitrary final velocity (or viceversa)

CLMVS – Solution of lab2

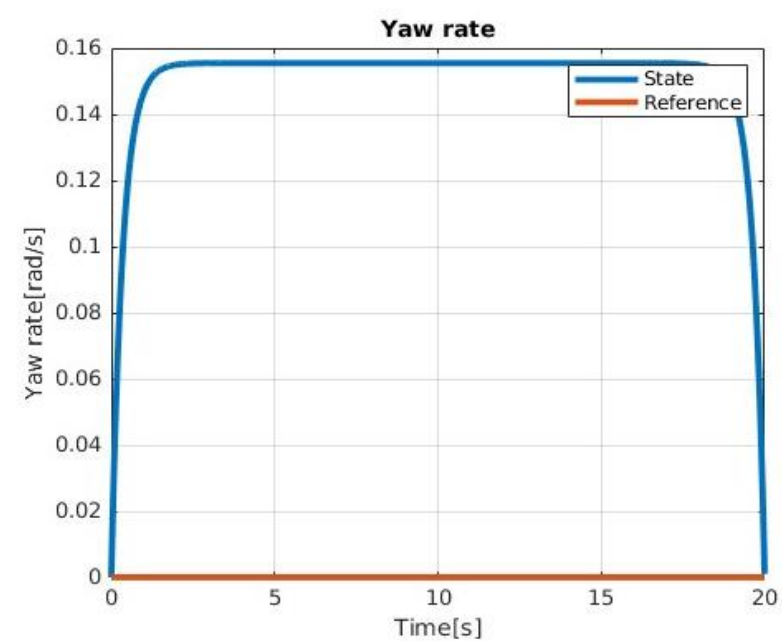
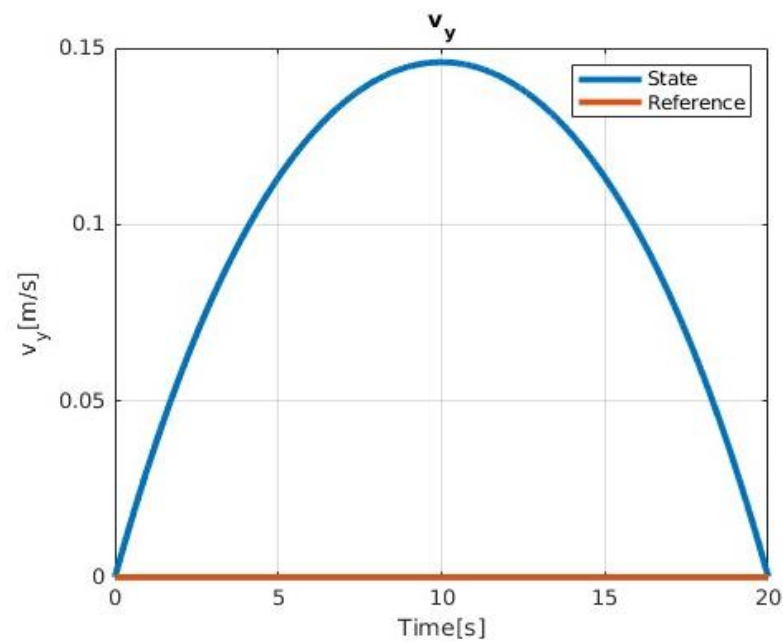
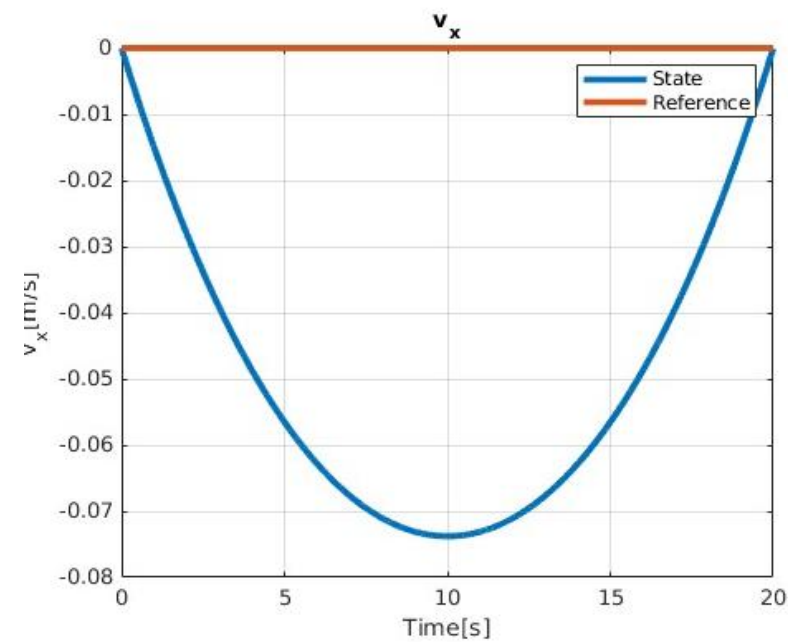
Controllability

- What happens if we still try to compute the system input from the controllability Gramian in the not fully controllable case (replacing $\text{inv}(G)$ with $\text{pinv}(G)$)?
- Expectations?





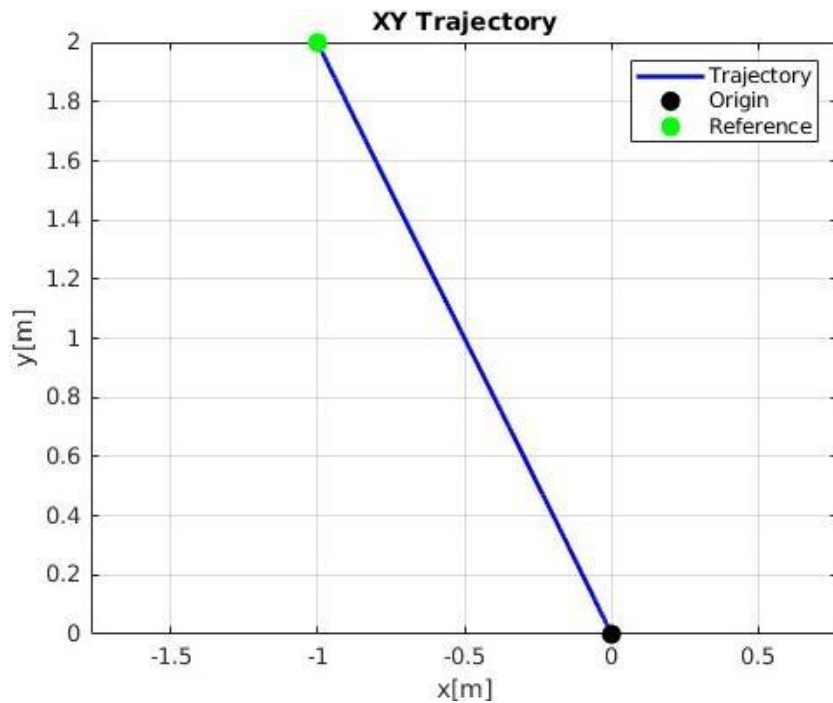
Compare to the controllable case



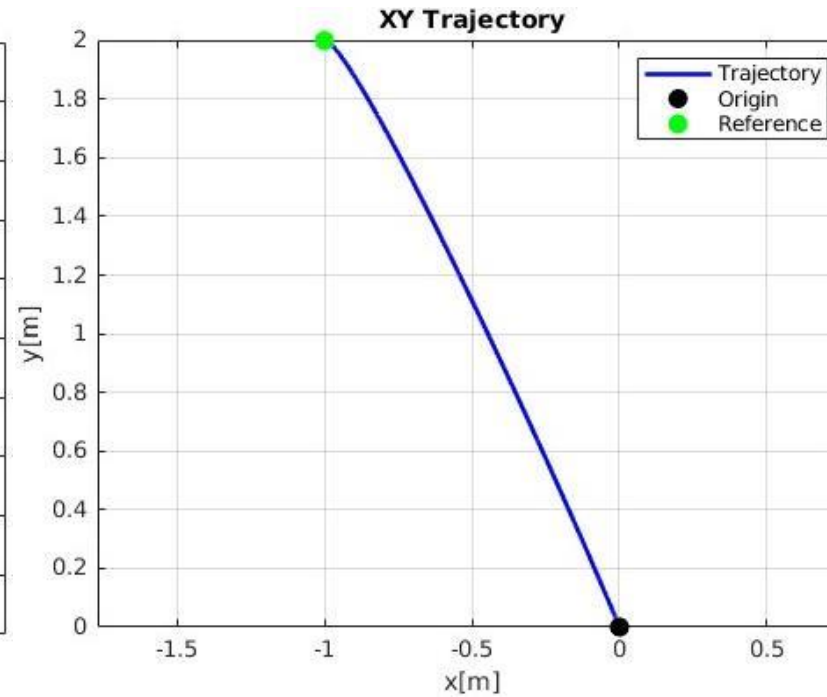
CLMVS – Solution of lab2

Controllability

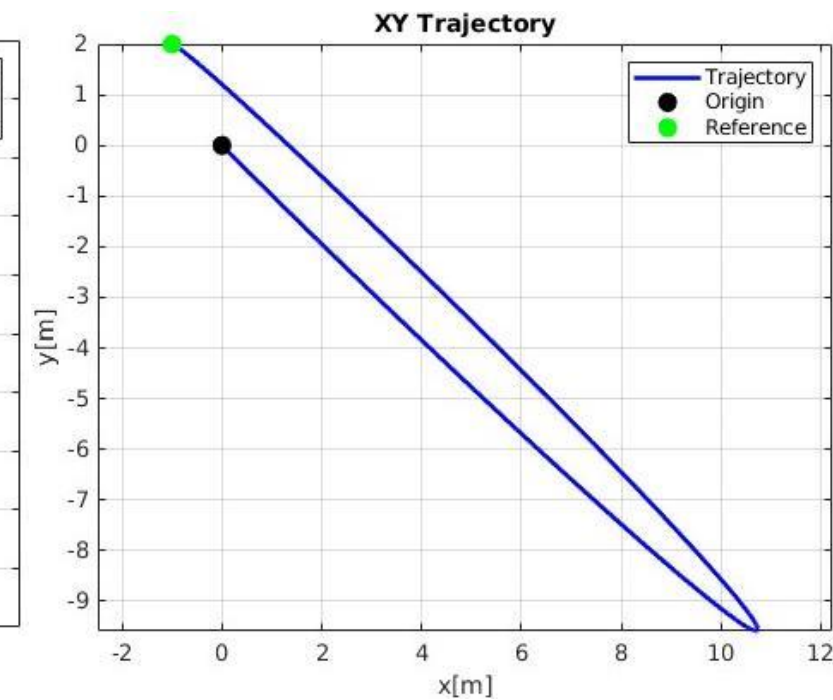
- Trajectory comparison:



Case 1



Case 2



Case 3

CLMVS – Solution of lab2

Controllability

- Numerically the final error (using $dt = 1e-3$) is:

$1.0e-03$ *

-0.0881
0.1585
0.1165
-0.0069
0.0123
0.1103

Case 1

$1.0e-03$ *

-0.0811
0.1614
0.1172
-0.0253
-0.0053
0.1480

Case 2

-0.0056
-0.0127
0.0001
-0.0809
-0.0798
0.0001

Case 3

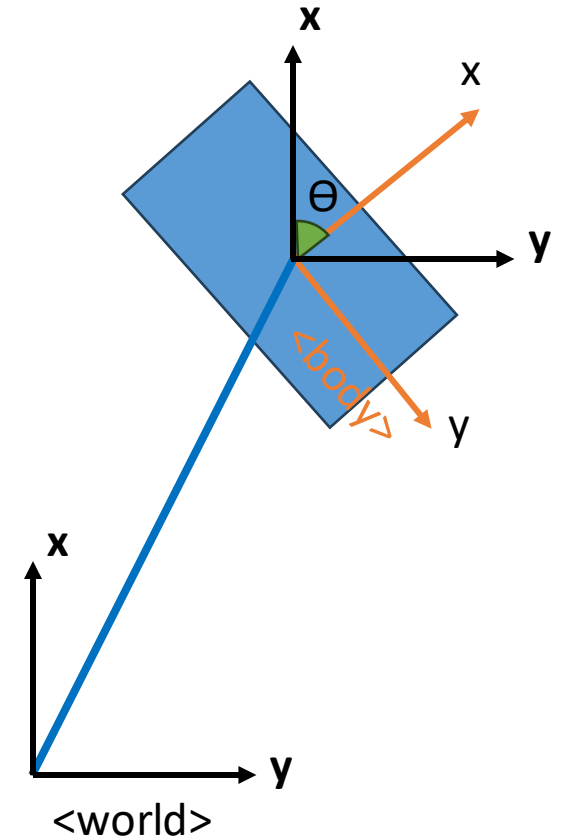
CLMVS – lab3

Observability

$$\begin{aligned} m\ddot{x} &= -\alpha\dot{x} + F_x \\ m\ddot{y} &= -\beta\dot{y} + F_y \\ J\dot{w} &= -\gamma w + N_z \end{aligned}$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\frac{d}{dt}x = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix} \begin{bmatrix} r_x \\ r_y \\ \theta \\ \dot{r}_x \\ \dot{r}_y \\ w \end{bmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/J \end{pmatrix} \begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix}$$



Computing Ko

Assume the positions are measured.

Since the observer is the positions and the orientation, which are coupled with their derivatives (the velocities), again we can deal with each two equation (position, velocity) as one independent system.

New state definition:

$$x \triangleq \begin{pmatrix} r_x \\ v_x \end{pmatrix} = \begin{pmatrix} x_1 \\ x_4 \end{pmatrix}$$

$$\begin{cases} \dot{r}_x = v_x \\ \dot{v}_x = -\frac{\alpha}{m} v_x + \frac{F_x}{m} \end{cases}$$

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\alpha/m \end{bmatrix} \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F_x$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_4 \end{pmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -\alpha/m \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Computing K_o

$$\dot{\hat{x}} = A\hat{x} + Bu_1 + K_{o_{surge}}(y - \hat{y}); K_{o_{surge}} = \begin{bmatrix} k_{x_1} \\ k_{x_4} \end{bmatrix}$$

$$\dot{x} = Ax + Bu_1$$

$$\tilde{x} = x - \hat{x};$$

$$\dot{\tilde{x}} = \dot{x} - \dot{\hat{x}} = Ax + Bu_1 - A\hat{x} - Bu_1 - K_{o_{surge}}(y - \hat{y})$$

$$\dot{\tilde{x}} = A\tilde{x} - K_{o_{surge}}C\tilde{x} = (A - K_{o_{surge}}C)\tilde{x}$$

We should choose K such that $(A - K_{o_{surge}}C)$ is asymptotically stable.

$$A - K_{o_{surge}}C = \begin{pmatrix} 0 & 1 \\ 0 & -\frac{\alpha}{m} \end{pmatrix} - \begin{pmatrix} k_{x_1} & 0 \\ k_{x_4} & 0 \end{pmatrix} = \begin{pmatrix} -k_{x_1} & 1 \\ -k_{x_4} & -\frac{\alpha}{m} \end{pmatrix}$$

Computing K_o

$$\det(\lambda I - [A - K_{o_{surge}} C]) = \begin{vmatrix} \lambda + k_{x_1} & -1 \\ k_{x_4} & \lambda + \frac{\alpha}{m} \end{vmatrix}$$

$$= \left(\lambda + \frac{\alpha}{m}\right)(\lambda + k_{x_1}) + k_{x_4}$$

$K_{o_{surge}}$

$$= \lambda^2 + \left(\frac{\alpha}{m} + k_{x_1}\right)\lambda + \left(k_{x_1}\frac{\alpha}{m} + k_{x_4}\right) = 0$$

$$\begin{cases} k_{x_1}\frac{\alpha}{m} + k_{x_4} = \omega_n^2 \\ \frac{\alpha}{m} + k_{x_1} = 2\zeta\omega_n \end{cases}$$

For our exercise we can choose $\zeta = 1$, $\omega_n = 1$ rad/s

Computing K_o

- We can repeat the same method for $[r_y, v_y]$, $[\theta, w]$ to compute all components of K_o

$$\bullet K_o = \begin{bmatrix} k_{x_1} & 0 & 0 \\ 0 & k_{x_2} & 0 \\ 0 & 0 & k_{x_3} \\ k_{x_4} & 0 & 0 \\ 0 & k_{x_5} & 0 \\ 0 & 0 & k_{x_6} \end{bmatrix}$$

$$\text{surge: } \begin{cases} k_{x_1} \frac{\alpha}{m} + k_{x_4} = w_n^2 \\ \frac{\alpha}{m} + k_{x_1} = 2\zeta w_n \end{cases}$$

$$\text{sway: } \begin{cases} k_{x_2} \frac{\beta}{m} + k_{x_5} = w_n^2 \\ \frac{\beta}{m} + k_{x_2} = 2\zeta w_n \end{cases}$$

$$\text{yaw: } \begin{cases} k_{x_3} \frac{\gamma}{J_w} + k_{x_6} = w_n^2 \\ \frac{\gamma}{J_w} + k_{x_3} = 2\zeta w_n \end{cases}$$

Exercise

You are given the template script *lab3.m*.

1. Implement a Luenberger observer, assuming that the state is observable, being the position variables only measured, tuning the observer gain matrix K_o . Assume that the initial state is known with error $[1, -1, 1.5, 0.5, -0.5, -1.5]$. Plot the difference between the true and estimated state.
2. Repeat 1 assuming that the whole state is measured.

